The Research of Customer Lifetime Value Related to Risk Factors in the Internet Business

Exampled by Online Car-hailing Industry

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***Abstract*—The quantitative method of Customer Lifetime Value is one of the most difficult problems in the field of Customer Relationship Management. The risk classification theory and emotion analysis method are introduced to modify the original model, and the prediction model of Customer Lifetime Value based on customer behavior analysis is built. In the empirical case of the car rental company, this model was trained with historical data and LightGBM algorithm. The final model is accurate and can also be interpreted according to the actual business. This paper provides an effective method of customer classification.**

Keywords—Customer Lifetime Value, Customer risk, Ride-hailing, LightGBM

I. INTRODUCTION

Enterprise managers have turned their attention from the market to the personal characteristics of customers, and take the customer as the center to adjust the company's strategic organization in the contemporary business field [1]. In the fiercely competitive environment, the loss of key customers is likely to cause significant effects in the company's cash flow. In-depth understanding of customer groups or individual customers will greatly affect the profitability of the company. Enterprise managers can use well-behaved prediction models to identify potential key users in their customer groups and prevent the loss of key customers through proper marketing measures. At the same time, it helps to allocate the company's limited marketing resources according to the contribution of customers to enterprise value creation [3].To quantify and evaluate the differences between customers, Customer Lifetime Value (CLV) is a widely recognized indicator.

According to the existing literature, CLV can be summarized as the monetary net present value of the profits that customers contribute to a company over its entire life cycle or a period of service. It can be used to identify the most profitable types of customers, predict the future profits that customers will contribute to the company, identify the most profitable customers and foster long-term relationships with customers. There are not a unified opinion of CLV evaluation methods in the academic field because of the changeable transaction field and complex business logic in the real world. Table I compares relevant representative literatures. Few studies have pointed out that the benefits brought by customers are accompanied by unpredictable risks until now, especially in the high-risk credit industry and leasing industry. Meanwhile, the incentive to customers and feedback will also bring word of mouth effect to the enterprise. This paper summarizes the research results of CLV, modifies and expands the traditional Customer Lifetime Value model through customer risk factors and emotional factors, and applies the model to the ride-hailing industry, which is in a social hot spot.

TABLE I. The Development History of CLV

References	Industry	Key conclusions
Dwyer (1997)	Insurance	Apply retention model and migration model to estimate CLV.
Kumar (2004)	Manufacturing	Build the random model of CLV based on allocation strategy.
Harsha (2008)	Credit	The model of CLV for bank risk based on the information of customer behavior is established.
Abdul (2011)	Retail	Evaluate CLV from customer and enterprise perspectives
Mohamed (2014)	Banking	Customers are subdivided based on the components of CLV
Maria (2018)	Telecom	Consider the variability of the customer lifecycle.

II. BACKGROUND AND CONCEPTUAL MODEL

A. Dominant Value

The dominant value of customers is the Customer Lifetime Value in the traditional sense. In the past literature, scholars mainly defined the customer life cycle value based on the profit present value of customer historical transactions and business experience to calculate the Customer Lifetime Value. Some scholars have subtracted various customer costs to maintain customer relationships when calculating CLV. In the research of this paper, we include factors such as customer loss and word-of-mouth effect into risk loss and potential value, etc. Therefore, the NPV model without other influencing factors was selected as the model basis. For example, Kumar proposed a universally recognized NPV calculation model of general CLV based on the permanent loss customer group [8], as shown in (1).

$$CLV_i = \sum_{t=0}^{\infty}\left[\frac{GC-M}{(1+d)^t}r^t\right]-A \tag{1}$$

GC is the average gross profit contribution, M is the marketing cost, A is the cost of acquiring customers, r is the customer retention rate and d is the discount rate.

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B. Potential Value

The potential value of customers refers to their own profitability if customers buy products or services from suppliers. Therefore, the potential value depends on the type of products or services which customers choose to some extent. In the estimation of potential value, we will consider customers' social characteristics, purchasing patterns in enterprises and types of products purchased as determinants of potential value. Using the profitability of each product and the purchase record, equation (2) can be obtained to calculate the Customer Potential Value (CPV) [12].

$$CPV_i = \sum_{k=1}^{K} prob_{i,k} \cdot profit_k \tag{2}$$

prop is the probability that customer *i* will buy portfolio *k*, and *profit* is the marginal profit of portfolio *k*.

C. Risk Loss

We calculate Customer Lifetime Value in terms of cash flow, excluding non-cash factors. This study focuses on the churn risk, default risk and operational risk involved by customers and enterprises.

Churn risk refers to the risk of losing cash flow due to customer loss. Causes of customer loss such as the customer's own cash flow cannot afford the consumer's spending in the enterprise. Or the products provided by enterprises are far from meeting the needs of customers due to market transformation. Considering the effect of individual Customer Churn rate, the measurement of Customer Churn Risk (CCR) is shown in (4) [13].

$$\text{Churn Rate} \qquad Churn(t) = h(t)e^{\overline{\alpha}\cdot\overline{N}} \tag{3}$$

$$\text{Churn Risk} \qquad CCR = \frac{\sum_{t=1}^{T} tChurn(t)}{\sum_{t=1}^{T} t} \tag{4}$$

Churn(t) represents the probability of customer loss at time *t*, *h(t)* is the basic risk function, and represents the instantaneous probability of customer loss, which can be constant or exponential distribution. $e^{\overline{\alpha}\cdot\overline{N}}$ is a covariate function. $\overline{N}$ is the influencing factor of objective reality. $\overline{\alpha}$ is the coefficient of the covariant. The meaning of (4) is to weight the loss risk of each period in the observation period. The closer the experiment is to the observation time, the more it can reflect the real loss of customers.

Default risk refers to the risk loss of cash flow caused by customers' failure to fulfill the contract due to personal willingness or objective conditions. In the case of customers with excellent credit, enterprises will make reasonable investment and resource allocation according to the cash flow to maximize the benefits. The expected loss of default risk can be expressed as the product of three different factors [14].

$$Expectedloss = PD \cdot LGD \cdot EAD \tag{5}$$

The Probability of Default (PD) represents the probability of payment failure. Loss Given Default (LGD) is the proportion of loan losses. Exposure at the Default (EAD) is the face value of an asset exposed to a specific default risk. In (5), PD is a parameter that cannot be directly obtained, and different industries have different calculation methods for it. The car-hailing industry has something in common with the bank credit business, and this study is based on the analysis of historical data. Therefore, we can refer to the CreditMetrics [14] model widely used in the banking industry. The experiment mainly uses the Value at Risk(VaR) of CreditMetrics model for calculation. VaR represents the maximum loss of a loan faces in future price fluctuations within a given confidence interval and holding period.

$$\begin{aligned} VaR_\alpha &= E(\omega) - \omega^* \\ &= E[\omega_0(1+r)] - \omega_0(1+r^*) \\ &= \omega_0[E(r) - r^*] \end{aligned} \tag{6}$$

E(ω) represents the expected value of the portfolio, ω represents the ending value of the portfolio, and ω^* represents the minimum ending value of the portfolio at confidence level *a*. ω_0 is the value of the asset portfolio at the beginning, and *r* is the return rate during the holding period. VaR can be regarded as expected loss, consistent with EAD. According to (5) and (6), the formula of calculating PD can be obtained, as shown in (7). Thus, we can also obtain the formula of Customer Default Risk(CDR), as shown in (8).

$$\begin{aligned} PD \cdot LGD \cdot EAD &= \omega_0[E(r) - r^*] \\ \Rightarrow PD &= \frac{E(r) - r^*}{LGD} \end{aligned} \tag{7}$$

$$CDR = \sum_{k=0}^{K} EAD_k[E(r_k) - r_k^*] \tag{8}$$

EAD refers to the face value of the assets exposed in the transaction, *K* is the number of credit products, *E(r)* is the expected rate of return of the portfolio during the holding period, and r^* is the lowest rate of return of the portfolio.

Operational risk refers to the risk of enterprise financial operation caused by unstable cash flow contributed by customers. In this study, the ratio between the standard value and the average value of the trading volume per unit time during the Customer observation period is used to represent the Customer Operational Risk(COR) [13].

$$COR = \frac{\dfrac{\sqrt{\sum_{t=1}^{T}(x_t - \overline{x})^2}}{T}}{\overline{x}} \tag{9}$$

t represents the trading time of the customer, *x* represents the trading transaction of the customer with the enterprise, *T* represents the observation time of the customer, $\overline{x}$ represents the average trading transaction of the customer in the

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observation period T. The closer the observation period is, the greater the weight of trading transaction fluctuation will be, and the greater the operational risk will be.

D. Sentiment Analysis

Customers' comments on products are the main channels for enterprises to receive customers' feedback directly, and they usually contain customers' own emotions. If companies can quickly analyze customer reviews and track product improvements, customer churn can be reduced.

E. Conceptual Model

Figure 1 illustrates the conceptual model of the CLV decision process in this paper. Four factors affecting customer value assessment are summarized. We can obtain data from the historical transactions, emotional feedback on products, customer credit conditions and so on.

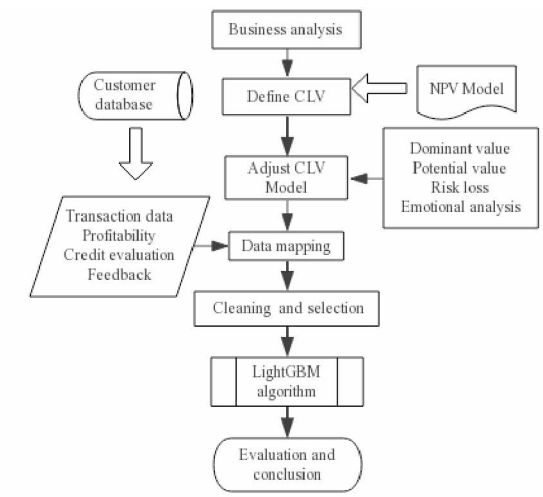

Fig.1. Flow chart of CLV modeling

III. Modeling Customer Lifetime Value

Initially, Customer Lifetime Value was a financial metric used to evaluate the present value of cash flows generated by customers minus the cost of customer acquisition and retention. Many studies have also proposed variations of CLV, as shown in (1). CLV is widely used and is a valuable indicator. However, this theory does not take into account other factors that affect customer value. Based on previous CLV studies, the value factors mentioned in our study are used to modify the most basic CLV model. The basic calculation model of CLV is shown in (10).

$$CLV_i = \sum_{t=1}^{h}\sum_{j=1}^{p}\left[\frac{GC_{i,j,t} - M_{i,j,t}}{(1+d^*)^t} - A_{i,j,t}\right] \tag{10}$$

h represents the customer's life cycle, p represents the number of products purchased by the customer, GC represents the marginal profit of the product, M represents the customer's marketing cost, A represents the customer's acquisition cost, and d^* represents the revised discount rate.

The measurement of potential value directly affects the measurement of customer value. Potential value calculation method is shown in (11). We get the predicted result of *prob* through customer feedback.

$$CPV_i = \sum_{t=1}^{h}\sum_{j=1}^{p} prob_{i,j,t} \cdot profit_{i,j,t} \tag{11}$$

The churn risk and operational risk were modified by applying the Beta risk discount rate proposed by Ryals [10], and default risk was directly applied in the CLV model. The emotional characteristics were judged by the SnowNLP algorithm, and the discount rate was modified in the form of risk correction. CDR is calculated by (8), and the modified customer discount rate is shown in (12) .

$$d^* = \overline{d} \cdot R_{Beta} = \frac{\overline{d} \cdot R}{\overline{R}} \tag{12}$$

d^* represents the discount rate of a customer's gross profit, $\overline{d}$ represents the average discount rate of all customers, R_{Beta} represents the Beta risk of the customer; R represents the overall risk of the customer, $\overline{R}$ represents the average total risk of all customers.

To sum up, we obtained a new CLV model. In order to distinguish it from the original CLV concept, this study named it as Comprehensive Customer Lifetime Value (CCLV). The calculation model of CCLV is shown in (13) . All the signs in this formula were given earlier.

$$\begin{aligned} CCLV_i &= CLV_i + CPV_i - CDR_i \\ &= \sum_{t=1}^{h}\sum_{j=1}^{p}\left[\frac{GC_{i,j,t} - M_{i,j,t}}{(1+d^*)^t} - A_{i,j,t}\right] \\ &\quad + \sum_{t=1}^{h}\sum_{j=1}^{p} prob_{i,j,t} \cdot profit_{i,j,t} \\ &\quad - \sum_{t=1}^{h}\sum_{j=1}^{p} EAD_{i,j,t}[E(r_{i,j,t}) - r_{i,j,t}^{\ *}] \end{aligned} \tag{13}$$

IV. A Case of the Online Car Rental Industry

A. Data

The data of this experiment comes from the historical database of a well-known domestic ride-hailing company. What we have obtained is customer transaction data for 3,246 customers from January 2018 to September 2018.

B. Algorithm

LightGBM algorithm was selected to realize Customer Lifetime Value classification modeling. Microsoft research Asia (MSRA) in 2017 launched a lightweight LightGBM Boosting framework. It can guarantee the accuracy of the model as well as the speed of calculation. In this study, numpy and sklearn, the third-party libraries of Python 3.6, were used for data processing.

C. Risk Measurement

1) Churn Risk

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In this paper, it is assumed that the rate of loss is constant. According to the actual needs of the business, the covariate factors can be determined as the proportion of average daily salary, the proportion of online hours, the proportion of daily orders received, the attendance rate, and the service score. Equation (3) is expanded into (14) in this section, and the problem is determined as parameter estimation for six parameters, $(\alpha_0,\alpha_1,\alpha_2,\alpha_3,\alpha_4,\alpha_5)$.

$$Churn(t) = \alpha_0 e^{\alpha_1 N_1 + \alpha_2 N_2 + \alpha_3 N_3 + \alpha_4 N_4 + \alpha_5 N_5} \quad (14)$$

Multivariate regression method and characteristic data of common and lost customers were used to fit the data.

TABLE II. The Estimation Results of Parameters

Covariate	α_0	α_1	α_2	α_3	α_4	α_5
Value	0.401	-0.387	-0.125	1.406	0.061	-0.539

From the parameter estimation results, the instantaneous customer loss rate is 0.401. The proportion of average daily wages can reflect the working efficiency of drivers to a certain extent. The proportion of daily orders plays a decisive role in all factors. The higher the value is, the higher the possibility of loss is. The more efficient the driver is, the more likely he will choose competitors for reasons such as rebate points, thus losing the driver. Through estimation model and formula, the probability value of each customer churn risk can be obtained.

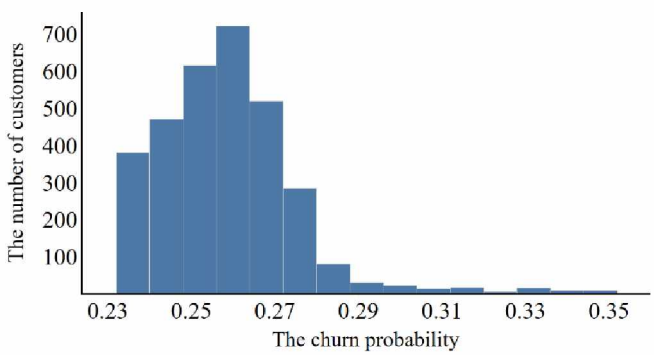

Fig.2. The distribution of customer churn probability

As can be seen from Figure 2, the risks of most customers are concentrated in the (0.24,0.28], accounting for 92.4% of the total number of customers. The number of customers in the (0, 0.24] accounts for 0.1%, the number of customers in the (0.24,0.3) accounts for 4.4%, and the number of customers between (0.3, 1) accounts for 3.3%. Based on the range distribution, we use 0.3 as critical point of customer churn. Analysis the loss probability of customers in (0.3, 1], most of these drivers' working experience is within three months, with unstable order quantity, low attendance rate and low customer stickiness, which are easy to be lost.

2) Default risk

EAD is calculated according to the vehicle type of the contract, and the expected rate of return and the lowest rate of return were determined according to different business types. The loss and theft of vehicles are not considered when making the decision of yield rate. The customer default risk distribution is shown in Figure 3.

The default risk of most customers is concentrated in the range of (0,10000), accounting for 83.7% of the total number of customers. Most customers' default risk is less than 20,000 yuan. When the driver defaults and still fails to make up the fee., the enterprise will take compulsory measures at the end of the next month. The amount of loss is about 2-3 months rental fee, vehicle service fee and other additional fees. The reason for the negative value in the figure is that the rental fee paid by the residual value of the vehicle and the driver is already greater than the cost of the contract. In this case, the driver will be in breach of the contract and no loss will be caused.

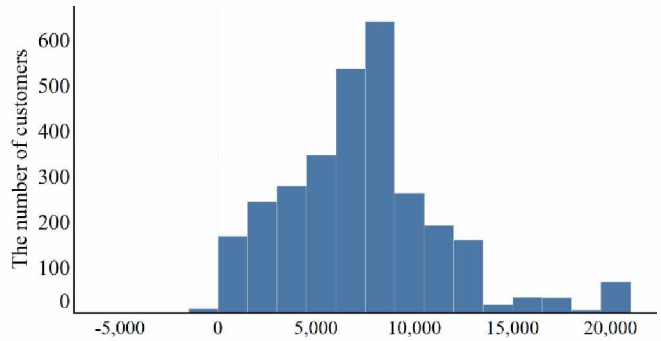

Fig.3. The distribution of customer default risk

3) Operational risk

Equation (9) is used to calculate the operational risk distribution histogram of operation risk, as shown in the Figure 4.The operational risk of most customers is concentrated in the range of (0,0.3), accounting for 91.2% of the total number of customers. Among them, the number of customers with operational risk distribution (0.25,0.3) increased slightly. These customers generally have long working hours, but their income is not stable enough, and some of them have income recession, which also represents a state of loss.

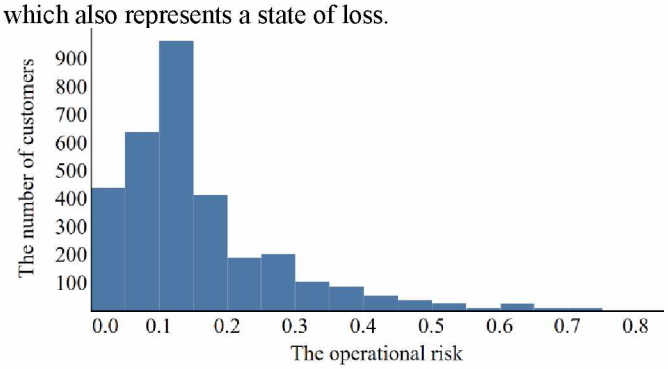

Fig.4. The distribution of customer operational risk

4) Emotional Factors Analysis

The emotional feedback of the customer group is mainly obtained through the regular telephone return visit of the enterprise sales staff, and the positive degree of the customer for the work will also be reflected in the conversation. SnowNLP is an emotion analysis library written in Python, which can handle Chinese text content conveniently. Through customers' emotional feedback, SnowNLP can give the emotional score between [0,1]. The closer the score is to 1, the more positive the emotional expression will be. Otherwise, the more negative the emotional expression will be. We classified the results into three categories, as shown in Table 3, the number of customers with

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positive emotions was about 4 times of the total number of customers with negative and neutral emotions. In this application, (1 - emotion score) is used as emotional risk value.

TABLE III. Distribution of Customer Emotional Types

Emotional Type	Positive	Neutral	Negative
Proportion	78.68%	10.97%	10.35%

5) Prediction of Customer Risk

The Bayesian network model of IBM's SPSS Modeler tool encapsulates the algorithms for building and learning. According to the risk values of churn risk, operational risk and emotional risk, the judgment result of whether a customer is a risk customer can be obtained through the customer samples. We use confidence, the probability that the customer is a risk customer, as the customer risk value. The risk value of the vast majority of customers is within the range [0,0.25], accounting for about 93.6% of all customers, and the risk value greater than 0.25 accounts for 6.4%.

D. The Modification of CLV Model

Apply (12) to calculate the customer's Beta risk, and the result is shown in Figure 5. The Beta risk of customers is between 0 and 2.4, which is suitable for the adjustment of discount rate. Among them, the number of customers with Beta risk greater than 1 is 1583, accounting for 48.8% of all selected customers.

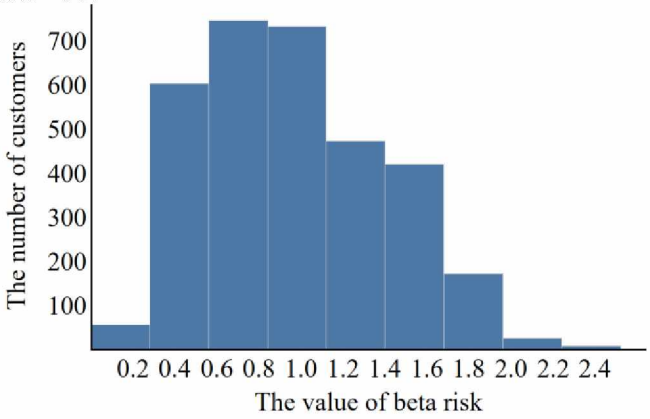

Fig.5.The distribution of Beta risk

The profit contribution shall be calculated according to the actual business of the company. The profitability value is set at 15% of the customer flow, which is the rental company's fee. Maintenance costs and acquisition costs are amortized according to actual company expenditures. The benchmark discount rate is set at 10%. In this study, CCLV was used to represent the risk-adjusted CLV value. Compare CCLV with CLV, that is, compare the calculation results of (1) and (15). Figure 6 shows the comparison between the CLV values of the top 20 customers calculated according to the CLV model and the calculation results of the modified CCLV model.

By comparing the two ranking situations of the top 20 customers, it was found that the 11th customer dropped from 11 to the 35th. All kinds of risk indicators of this customer were queried, as shown in Table IV. Compared with the risk value of overall customers, this customer has higher emotional risk and default risk. Default risk can directly affect the total value of customers. Customers with fluctuating values are also likely to cause great losses to enterprises.

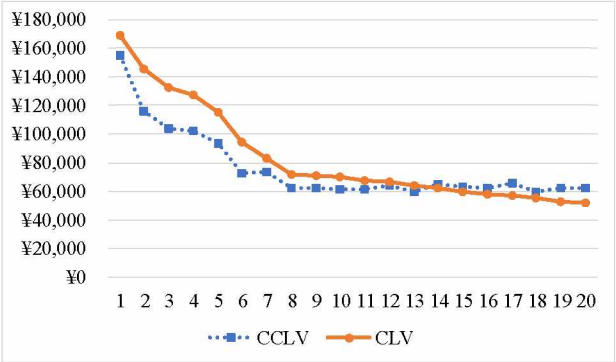

Fig.6. CLV and CCLV for the top 20 customers

TABLE IV. The various risk values of the 11th customer

The type of risk	Churn risk	Default risk	Operational risk	Emotional risk
Value	0.26	12000	0.11	0.96
Level	Medium	High	Low	High

E. Customer Classification Based on LightGBM Algorithm

Based on the actual business revenue needs, we divide customers into three categories in terms of CCLV, (-∞,5000], (5000,20000) and [20000, + ∞). LightGBM algorithm was used to input the social feature information, feedback information, contract information, credit status, vehicle operation and other data which can be directly available in the enterprise database of 3246 customers. The grid search method and 5-fold cross-validation were used to obtain the best parameters of the model, train the customer classification model, and compare the performance of different estimator types, as shown in Table V. Among the four estimator types, GOSS and DART perform better. Considering the factors of overfitting, GOSS consumes less. Therefore, GOSS is adopted as the final model. The top 10 features in importance are shown in Figure 7.

TABLE V. Best Parameters for Different Estimators

Estimators	n_estimators	num_leaves	max_depth	Accuracy
GBDT	120	20	7	96.61%
RF	100	20	9	94.56%
DART	100	35	9	96.71%
GOSS	100	25	7	96.82%

As the most direct distinguishing factor of customer value, gross profit occupies the first place of feature importance. Secondly, the profitability of each month and the contract rental payment represent the potential value and default risk respectively, which are also two important factors. The number of customer ride-hailing periods represent the customer operating cycle. The longer the cycle, the more adverse to the operation of enterprise capital, the greater the risk. Telephone feedback represents the customer's emotional factors, while points represent the customer's user stickiness and activity. The

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feature ranking basically conforms to the actual business situation.

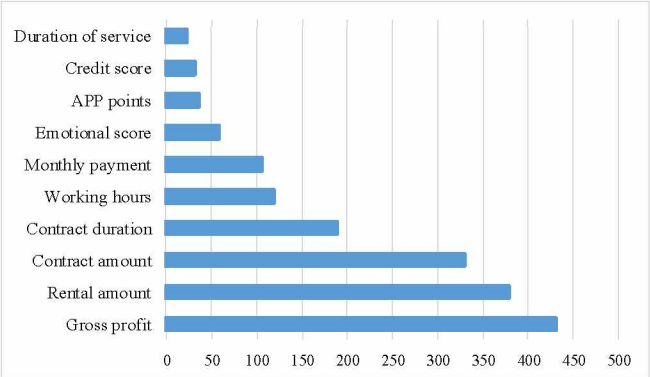

Fig.7. Chart of feature importance ranking

V. Conclusion

On the basis of previous studies, this paper proposes the concept of CCLV, classifies risk factors, and proposes quantitative methods for different risk factors. In addition to customer profit, this model takes into account customer social characteristics, personal default risk, potential profitability and emotional factors, provides a method to calculate customer value, and builds a value prediction model based on customer data, expanding the application field of LightGBM.

There are still two shortcomings in this study: first, the classification of customer risk is only based on the analysis of enterprise business needs and past literatures, which may lead to the omission of risk factors; The other is the selection of prediction indicators for the CCLV model. Due to the enterprise thinking of special business processes, there are certain subjective thoughts. In the future research, the classification of customer risk and the universality of indicators are two directions worth exploring.

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